# Math 110 Review

## Solutions

## §2 Problem set

### §2.1 Minimal polynomial (HW7)

**HW7 P1.**

**Idea.** *$A^2 = -I$ and $(B-I)^2 = 0$ give low-degree polynomials that kill $A$ and $B$; then rule out degree 1.*

(a) $A^2 = -I$, so $m_A \mid x^2+1$. $A$ is not a multiple of $I$ ($Ae_1 = e_2$ is not a multiple of $e_1$), so $\deg m_A \neq 1$. Hence $m_A = x^2+1$ over $\mathbb{R}$. Over $\mathbb{C}$ the definition (first $k$ with $A^k \in \operatorname{span}(I, \dots, A^{k-1})$) gives the same polynomial: $x^2+1 = (x-i)(x+i)$.

(b) $(B-I)^2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)^2 = 0$ and $B \neq I$, so $m_B = (x-1)^2$.

(c) $A$ over $\mathbb{R}$: $x^2+1$ has no real roots, so it is not a product of real linear factors: not diagonalizable. Over $\mathbb{C}$: distinct roots $\pm i$, diagonalizable. $B$: repeated root, not diagonalizable. This matches Homework 6.

**HW7 P2.**

**Idea.** *$D$ lowers degree, so a power of it is 0; find the first one.*

(a) $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$, while $D^3 x^3 = 6 \neq 0$. So $m_D \mid x^4$ and is not $x^k$ for $k \leq 3$: $m_D = x^4$. (b) Repeated root: not diagonalizable. (Also $E(0, D) =$ constants, of dimension $1 < 4$.)

**HW7 P3.**

(a) **Idea.** *If $a_0 = 0$, then $x$ divides $m_T$; since $T$ is invertible, the factor $T$ can be cancelled, leaving a smaller polynomial that kills $T$.*
If $a_0 = 0$, then $m_T = xq$ with $\deg q = d-1$, and $0 = m_T(T) = Tq(T)$. Multiply by $T^{-1}$: $q(T) = 0$, contradicting minimality of $m_T$ (if $d = 1$, $q = 1$ and $q(T) = I \neq 0$).

(b) **Idea.** *Move the constant term to one side and factor $T$ out of the rest.*
From $m_T(T) = 0$: $T\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big) = -a_0 I$. So $T^{-1} = -\frac{1}{a_0}\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big)$.

(c) **Idea.** *Apply (b), then factor $m_T$ to read off the eigenvalues.*
$a_0 = 4$, so $T^{-1} = -\frac{1}{4}(T^3 + T^2 - T - 5I)$. $m_T = (x-1)^2(x^2+3x+4)$ (check: 1 is a root of $m_T$ and of $m_T' = 4x^3 + 3x^2 - 2x - 5$). The eigenvalues are 1 and $\frac{-3 \pm i\sqrt{7}}{2}$. The root 1 is repeated, so $T$ is not diagonalizable.

**HW7 P4.**

**Idea.** *Substituting $x \mapsto x + \lambda$ (or $x \mapsto x/c$) turns polynomials that kill $T$ into polynomials that kill $T - \lambda I$ (or $cT$), and back. So the two minimal polynomials divide each other's transforms and have the same degree.*

(a) $p(x) = m_T(x+\lambda)$ is monic of degree $d = \deg m_T$ and $p(T - \lambda I) = m_T(T) = 0$, so $m_{T-\lambda I} \mid p$ and $\deg m_{T-\lambda I} \leq d$. Likewise $r(x) = m_{T-\lambda I}(x-\lambda)$ is monic with $r(T) = m_{T-\lambda I}(T - \lambda I) = 0$, so $d \leq \deg m_{T-\lambda I}$. A monic divisor of $p$ of the same degree is $p$.

(b) $p(x) = c^d m_T(x/c)$ is monic of degree $d$ (leading coefficient $c^d \cdot c^{-d}$) and $p(cT) = c^d m_T(T) = 0$, so $m_{cT} \mid p$. The same argument with $cT$ and $1/c$ gives $d \leq \deg m_{cT}$. So $m_{cT} = p$.

**HW7 P5.**

**Idea.** *The roots of $m_A$ are the eigenvalues and $\deg m_A \leq 2$, which leaves two candidates.*

$\deg m_A \leq 2$ (Cayley–Hamilton) and its only root is $\lambda$, and over $\mathbb{C}$ it splits, so $m_A = x - \lambda$ or $(x-\lambda)^2$. $m_A = x - \lambda$ exactly when $A - \lambda I = 0$. $A$ is diagonalizable iff $m_A$ has no repeated root iff $m_A = x - \lambda$ iff $A = \lambda I$.

**HW7 P6.**

**Idea.** *Triangular, so the eigenvalues are on the diagonal; list the candidate minimal polynomials and test them.*

Eigenvalues $2, 3$. $E(2) = \operatorname{null}\left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = \operatorname{span}(e_1)$, $E(3) = \operatorname{null}\left(\begin{smallmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) = \operatorname{span}(e_3)$. Candidates (roots exactly $2, 3$, degree $\leq 3$): $(x-2)(x-3)$, $(x-2)^2(x-3)$, $(x-2)(x-3)^2$. $(C-2I)(C-3I)$ and $(C-2I)(C-3I)^2$ have $(1,2)$ entry $\mp 1 \neq 0$; $(C-2I)^2 = \operatorname{diag}(0,0,1)$ and $(C-2I)^2(C-3I) = 0$. So $m_C = (x-2)^2(x-3)$: not diagonalizable ($\dim E(2) + \dim E(3) = 2 < 3$).

**HW7 P7.**

**Idea.** *Lagrange interpolation: a product of factors $(x - \lambda_k)$ vanishes at every $\lambda_k$ but $\lambda_j$; divide by its value at $\lambda_j$.*

$p_j(x) = \prod_{k \neq j} \frac{x-\lambda_k}{\lambda_j - \lambda_k}$. Since $m_T$ has distinct roots, $T$ is diagonalizable and $V = \bigoplus_k E(\lambda_k, T)$. For $v \in E(\lambda_k, T)$, $p_j(T)v = p_j(\lambda_k)v$, which is $v$ if $k = j$ and 0 otherwise. So for $v = \sum_k v_k$, $p_j(T)v = v_j$: the projection onto $E(\lambda_j, T)$ along $\bigoplus_{k \neq j} E(\lambda_k, T)$.

**M1.**

**Idea.** *Find a low-degree polynomial relation, then rule out its proper divisors. In (c), $B$ is already in Jordan form, so read $m_B$ off the blocks: the largest block for each eigenvalue gives its exponent.*

(a) Every entry of $A^2$ is 3, so $A^2 = 3A$ and $m_A \mid x(x-3)$. $A \neq 0$ and $A \neq 3I$, so $m_A = x(x-3)$. Distinct roots: diagonalizable.

(b) $Pe_1 = e_2$, $Pe_2 = e_3$, $Pe_3 = e_1$, so $P^3 = I$ and $m_P \mid x^3 - 1$. Since $e_1, Pe_1, P^2e_1$ is a basis, no nonzero polynomial of degree $\leq 2$ kills $e_1$, so $m_P = x^3 - 1 = (x-1)(x^2+x+1)$. Over $\mathbb{C}$ the roots $1, \omega, \bar{\omega}$ are distinct: diagonalizable. Over $\mathbb{R}$, $x^2 + x + 1$ has no real root: not diagonalizable.

(c) $B = J_2(3) \oplus J_1(3) \oplus J_1(5)$, so $m_B = \operatorname{lcm}\big((x-3)^2, x-3, x-5\big) = (x-3)^2(x-5)$. Repeated root: not diagonalizable.

**M2.**

(a) **Idea.** *Test the basis vectors: $x^k$ is an eigenvector.*
$T(x^k) = kx^k$, so $1, x, \ldots, x^n$ is an eigenbasis with distinct eigenvalues $0, 1, \ldots, n$. $T$ is diagonalizable and $m_T = x(x-1)\cdots(x-n)$.

(b) **Idea.** *$S - I$ is a "difference operator" that lowers degree, like differentiation; so it is nilpotent, and the question is the first power that kills everything.*
$(S-I)p = p(x+1) - p(x)$: the leading terms cancel, so deg drops by at least 1 and constants go to 0. Hence $(S-I)^{n+1} = 0$, so $m_S \mid (x-1)^{n+1}$. Also $(S-I)x^k = (x+1)^k - x^k = kx^{k-1} + (\text{lower})$, so $(S-I)^n x^n = n! \neq 0$. So $m_S = (x-1)^{n+1}$, a repeated root: not diagonalizable.

**M3.**

**Idea.** *$T^3 = T$ says $x^3 - x$ kills $T$; factor it.*

$m_T \mid x^3 - x = x(x-1)(x+1)$, so $m_T$ is a product of distinct real linear factors and $T$ is diagonalizable. Its eigenvalues are roots of $m_T$, hence in $\{-1, 0, 1\}$.

**M4.**

**Idea.** *A monic polynomial of degree 1 is $x - c$, and $m_T = x - c$ says $T - cI = 0$.*

If $T = cI$, then $x - c$ kills $T$ and $\deg m_T \geq 1$ (as $V \neq 0$), so $m_T = x - c$. Conversely, if $m_T = x - c$, then $T - cI = m_T(T) = 0$.

**M5.**

**Idea.** *Substitute $x \mapsto 1/x$ in $m_T(T) = 0$ and clear denominators by multiplying by $T^{-d}$.*

$q(x) = \frac{1}{a_0} x^d m_T(1/x) = \frac{1}{a_0}\big(1 + a_{d-1}x + \cdots + a_0 x^d\big)$ is monic of degree $d$ ($a_0 \neq 0$ by HW7 P3). $q(T^{-1}) = \frac{1}{a_0} T^{-d} m_T(T) = 0$, so $m_{T^{-1}} \mid q$ and $\deg m_{T^{-1}} \leq d$. Applying this to $T^{-1}$ in place of $T$ gives $d \leq \deg m_{T^{-1}}$. So $m_{T^{-1}} = q$.

**M6.**

**Idea.** *Let $r_i$ be the step where the chain* $\operatorname{null}(T-\lambda_i I)^k$ *reaches $G(\lambda_i, T)$. $\prod(x-\lambda_j)^{r_j}$ kills every $G$, giving $k_i \le r_i$. For $\ge$: if $k_i < r_i$, some vector of $G(\lambda_i, T)$ survives $(T-\lambda_i I)^{k_i}$, and the other factors of $m_T$ are injective on $G(\lambda_i, T)$, so $m_T(T)$ would not kill it.*

Let $r_i$ be the smallest $k$ with $\operatorname{null}(T-\lambda_i I)^k = G(\lambda_i, T)$.

1. $q = \prod_j (x-\lambda_j)^{r_j}$ kills each $G(\lambda_j, T)$ (factors commute), and these span $V$ (Prop. 73; by (T9) the $\lambda_j$ are all the eigenvalues of $T$). So $m_T \mid q$ and $k_i \le r_i$.
2. For $j \ne i$, $T-\lambda_j I$ is injective on $G(\lambda_i, T)$: a vector it kills is an eigenvector for $\lambda_j$ lying in $G(\lambda_i, T)$, and generalized eigenvectors for distinct eigenvalues are independent (Prop. 72), so it is 0.
3. If $k_i < r_i$, pick $v \in G(\lambda_i, T)$ with $w = (T-\lambda_i I)^{k_i} v \ne 0$; $w \in G(\lambda_i, T)$ (invariant). Then $0 = m_T(T)v = \prod_{j \ne i}(T-\lambda_j I)^{k_j} w \ne 0$ by step 2. Contradiction, so $k_i = r_i$.

In a Jordan basis, $\operatorname{null}(T-\lambda_i I)^k = G(\lambda_i, T)$ exactly when $k$ is at least the largest block size for $\lambda_i$.

**M7.**

**Idea.** *Block-count table: differences of the $d_k$ count blocks of size $\ge k$.*

$d_0, \dots, d_5 = 0, 3, 5, 6, 7, 7$ ($d_5 = 7$ since $\dim V = 7$). Blocks of size $\ge 1$: 3; $\ge 2$: 2; $\ge 3$: 1; $\ge 4$: 1; $\ge 5$: 0. Sizes $4, 2, 1$ (sum 7). $T \sim J_4(4) \oplus J_2(4) \oplus J_1(4)$, $m_T = (x-4)^4$, $\chi_T = (x-4)^7$.

**M8.**

**Idea.** *Translate:* $\dim E(1)$ *is the number of blocks for 1; the exponent in $m_T$ is the largest block; the total dimension fixes the rest.*

Eigenvalue 1: one block, of size 2. Eigenvalue $-2$: blocks of size 1, total $4-2=2$. So $T \sim J_2(1) \oplus J_1(-2) \oplus J_1(-2)$, $\chi_T = (x-1)^2(x+2)^2$, $\operatorname{tr} T = 1+1-2-2 = -2$.

**M9.**

**Idea.** *$m_A(A) = 0$ is a rule for rewriting $A^2$ in terms of $A$ and $I$; use it to reduce any power, and solve it for $A^{-1}$.*

$A$ is not scalar, so $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A = x^2 - 5x - 2$. From $A^2 - 5A = 2I$: $A(A-5I) = 2I$, so $A^{-1} = \frac{1}{2}(A-5I) = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$. $A^3 = A(5A+2I) = 5A^2 + 2A = 5(5A+2I) + 2A = 27A + 10I$.

## §2.2 Generalized eigenvectors and Jordan form (HW8)

**HW8 P1.**

(a) **Idea.** *Differentiation lowers degree, so enough derivatives kill any polynomial; the index is the first power that kills $x^n$, the slowest vector to die.*
$D$ lowers degree by 1 and kills constants, so $D^{n+1}p = 0$ for every $p \in \mathcal{P}_n(\mathbb{C})$, while $D^n x^n = n! \ne 0$. So $D$ is nilpotent of index $n+1$.

(b) $E(0, D) = \operatorname{null} D =$ constants, so the geometric multiplicity is 1. $G(0, D) = \mathcal{P}_n(\mathbb{C})$ since $D$ is nilpotent, so the algebraic multiplicity is $n+1$.

(c) **Idea.** *One Jordan block means one chain. Start from the vector that takes longest to die ($x^n$) and differentiate repeatedly; dividing by factorials makes each step land on the next basis vector with coefficient 1.*
$f_k = x^k/k!$ for $k = 0, \dots, n$: $Df_0 = 0$ and $Df_k = x^{k-1}/(k-1)! = f_{k-1}$. In the basis $f_0, f_1, \dots, f_n$, column $k$ of the matrix has a single 1 in row $k-1$: the matrix is the single block $J_{n+1}(0)$.

**HW8 P2.**

(a) **Idea.** *Translate the data into blocks: geometric multiplicity = number of blocks, algebraic = total size, largest block = exponent in $m$.*
Number of blocks = geometric multiplicity, total size = algebraic multiplicity: $J_3(2) \oplus J_1(-1) \oplus J_1(-1)$,

$$\begin{pmatrix} 2&1&0&0&0 \\ 0&2&1&0&0 \\ 0&0&2&0&0 \\ 0&0&0&-1&0 \\ 0&0&0&0&-1 \end{pmatrix}, \qquad m = (x-2)^3(x+1).$$

(b) Two blocks for 2 with total size 3: $J_2(2) \oplus J_1(2) \oplus J_1(-1) \oplus J_1(-1)$, i.e. the matrix in (a) with the $(2,3)$ entry changed to 0. The largest block for 2 has size 2, so $m = (x-2)^2(x+1)$.

**HW8 P3.**

(a) $N = J - I$ has $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$. So $\operatorname{null} N = \operatorname{span}(e_1)$; $N^2 e_3 = e_1$ and $N^2 e_1 = N^2 e_2 = 0$, so $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2)$; $N^3 = 0$, so $\operatorname{null} N^3 = \mathbb{F}^3$.

(b) $G(1, J) = \operatorname{null}(J - I)^3 = \mathbb{F}^3$. Geometric multiplicity $\dim \operatorname{null}(J - I) = 1$; algebraic multiplicity $\dim G(1, J) = 3$.

(c) **Idea.** *Split $J$ into a scalar part and a nilpotent part that commute, then use the binomial theorem; it stops at $N^2$ because $N^3 = 0$.*

$J = I + N$ with $IN = NI$ and $N^3 = 0$, so $J^m = I + mN + \binom{m}{2}N^2 = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$ (with $\binom{m}{2} = \frac{m(m-1)}{2}$; for $m = 1$ this is $J$).

**HW8 P4.**

(a) **Idea.** *Null spaces of a block diagonal matrix add block by block, and for one block $J_j(0)^k$ is a shift by $k$ places.* For one block $J_j(0)$, $\dim \operatorname{null} J_j(0)^k = \min(k, j)$. Add over the blocks of sizes 3 and 2: $k = 0$: 0; $k = 1$: $1+1 = 2$; $k = 2$: $2 + 2 = 4$; $k = 3$: $3 + 2 = 5$; $k = 4$: $3 + 2 = 5$.

(b) **Idea.** *Differences of the $d_k$ count blocks of size $\geq k$; the total dimension pins down the largest size.* The number of blocks of size $\geq k$ is $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$: $2 - 0 = 2$ blocks of size $\geq 1$, $4 - 2 = 2$ of size $\geq 2$, $5 - 4 = 1$ of size $\geq 3$. So there are two blocks, both of size at least 2, exactly one of size at least 3. The three numbers alone do not fix the larger size ($J_4(0) \oplus J_2(0)$ on $\mathbb{C}^6$ also gives $2, 4, 5$), so we use that $N$ acts on the same 5-dimensional space as in (a): then $\operatorname{null} N^3 = V$ and $N^3 = 0$, so no block has size above 3. Hence the sizes are 3 and 2. (Equivalently: the sizes are 2 and $s \geq 3$, and $2 + s = \dim V = 5$.)

**HW8 P5.**

**Idea.** *For each eigenvalue, make the table of $d_k$ and take differences (Prop. 77).*

Eigenvalue 1: $d_1, d_2, d_3 = 1, 2, 2$, so 1 block of size $\geq 1$, 1 of size $\geq 2$, none of size $\geq 3$: $J_2(1)$. Eigenvalue 2: $d_1, \ldots, d_4 = 2, 3, 4, 4$, so 2 blocks of size $\geq 1$, 1 of size $\geq 2$, 1 of size $\geq 3$, none of size $\geq 4$: $J_3(2)$ and $J_1(2)$. Check: $2 + 3 + 1 = 6$. So $T \sim J_2(1) \oplus J_3(2) \oplus J_1(2)$.

**HW8 P6.**

(a) $\chi = (x - 4)^5(x + 1)$. (b) $m = (x - 4)^3(x + 1)$ (the largest block for 4 has size 3). (c) Eigenvalue 4: geometric 2 (two blocks), algebraic 5; eigenvalue $-1$: geometric 1, algebraic 1. (d) Not diagonalizable: $m$ has the repeated root 4 (equivalently, $2 \neq 5$).

**HW8 P7.**

**Idea.** *Everything about a block diagonal matrix splits into blocks, so count block by block; for one block the null space of $J_j(0)^k$ is easy because $J_j(0)^k$ is a shift.*

Choose a basis in which $N$ is in Jordan form with blocks $J_{j_1}(0), \ldots, J_{j_r}(0)$. $N^k$ is block diagonal with blocks $J_{j_i}(0)^k$, so $\dim \operatorname{null} N^k = \sum_i \dim \operatorname{null} J_{j_i}(0)^k$. On one block with basis $e_1, \ldots, e_j$, $J_j(0)^k e_l = e_{l-k}$ for $l > k$ and 0 for $l \leq k$; so $\operatorname{null} J_j(0)^k = \operatorname{span}(e_1, \ldots, e_{\min(k,j)})$ and $\dim \operatorname{null} J_j(0)^k = \min(k, j)$. Then $\min(k, j) - \min(k - 1, j)$ is 1 if $j \geq k$ and 0 if $j < k$. Summing over the blocks, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ counts the blocks with $j_i \geq k$.

**G1.**

**Idea.** *Get the eigenvalues from a characteristic polynomial, then make the $d_k$ table, then build chains from the top.*

The upper-left block has trace 4 and determinant 4, so $\chi_A = (x - 2)^2 \cdot (x - 2) = (x - 2)^3$: only eigenvalue 2. $A - 2I = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ has rank 1, so $E(2, A) = \{y = x\} = \operatorname{span}\big((1, 1, 0), (0, 0, 1)\big)$ and $d_1 = 2$. $(A - 2I)^2 = 0$, so $d_2 = 3$ and $G(2, A) = \mathbb{C}^3$. Blocks: 2 of size $\geq 1$, 1 of size $\geq 2$: $J_2(2) \oplus J_1(2)$, $m_A = (x - 2)^2$. Jordan basis: top $e_1 \notin \operatorname{null}(A - 2I)$, bottom $(A - 2I)e_1 = (-1, -1, 0)$; then the eigenvector $e_3$, independent of that bottom. In the basis $(-1, -1, 0), e_1, e_3$ the matrix is $J_2(2) \oplus J_1(2)$ (check: $Ae_1 = (1, -1, 0) = 2e_1 + (-1, -1, 0)$).

**G2.**

**Idea.** *Same as for eigenspaces: $S$ commutes with every polynomial in $T$, so it preserves null spaces of polynomials in $T$.*

$G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$. If $(T - \lambda I)^n v = 0$, then $(T - \lambda I)^n Sv = S(T - \lambda I)^n v = 0$.

**G3.**

(a) **Idea.** *$(\lambda I + N)v = 0$ says $v$ is an eigenvector of $N$ for $-\lambda$, and 0 is the only eigenvalue of $N$.*
If $\lambda \neq 0$ and $(\lambda I + N)v = 0$, then $Nv = -\lambda v$; since $-\lambda$ is not an eigenvalue of $N$, $v = 0$. So $\lambda I + N$ is injective, hence invertible. If $\lambda = 0$, $N$ is not injective ($V \neq 0$ and $N^k = 0$). (Explicitly, $(\lambda I+N)^{-1} = \lambda^{-1}\sum_{j=0}^{n-1}(-N/\lambda)^j$.)

(b) **Idea.** *With one eigenvalue there is one generalized eigenspace, and it is everything.*
By Prop. 73, $V = G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, so $(T - \lambda I)^n = 0$.

**G4.**

**Idea.** *Block sizes add to the algebraic multiplicity 4; the largest is the exponent 2 in $m_T$.*

Sizes with largest 2 and sum 4: $\{2, 2\}$ or $\{2, 1, 1\}$. So $J_2(2) \oplus J_2(2)$ or $J_2(2) \oplus J_1(2) \oplus J_1(2)$. They have the same $\chi_T$ and $m_T$; $\dim E(2, T)$ (2 versus 3) tells them apart.

**G5.**

**Idea.** *Over $\mathbb{C}$, diagonalizable means $E(\lambda, T) = G(\lambda, T)$ for every $\lambda$.*

If $0 \neq v \in G(\lambda, T)$, the hypothesis gives $Tv = \mu v$ for some $\mu$; then $0 = (T - \lambda I)^n v = (\mu - \lambda)^n v$, so $\mu = \lambda$. Hence $G(\lambda, T) \subseteq E(\lambda, T)$, so $G(\lambda, T) = E(\lambda, T)$ for each eigenvalue. Then $V = \bigoplus G(\lambda_i, T) = \bigoplus E(\lambda_i, T)$ (Prop. 73), and a union of bases of the eigenspaces is an eigenbasis.

## §2.3 Trace (HW9)

**HW9 P1.**

**Idea.** *Trace can be computed in any basis, so use the one where $P$ is simplest: $P$ is 0 on $\operatorname{null} P$ and the identity on $\operatorname{range} P$.*

$V = \operatorname{null} P \oplus \operatorname{range} P$. Take a basis $u_1, \ldots, u_a$ of $\operatorname{null} P$ and $w_1, \ldots, w_b$ of $\operatorname{range} P$. $Pu_i = 0$, and $w_j = Pv$ for some $v$ gives $Pw_j = P^2 v = Pv = w_j$. So in the basis $u_1, \ldots, u_a, w_1, \ldots, w_b$, $\mathcal{M}(P) = \operatorname{diag}(0, \ldots, 0, 1, \ldots, 1)$ with $b$ ones, and $\operatorname{tr} P = b = \dim \operatorname{range} P = \operatorname{rank} P$.

**HW9 P2.**

(a) **Idea.** *$m_T$ fixes the largest block for each eigenvalue and $\dim V = 5$ fixes the total size; list the ways to fill 5.*
The largest block for 2 has size 2 and the largest block for $-1$ has size 1. The eigenvalues of $T$ are exactly the roots $2, -1$ of $m_T$, so with $a = \dim G(2, T) \geq 2$ and $b = \dim G(-1, T) \geq 1$, Prop. 73 gives $a + b = 5$: $(a, b) \in \{(2, 3), (3, 2), (4, 1)\}$.

(b) $\operatorname{tr} T = 2a - b$: 1, 4, 7.

(c) $\operatorname{tr} = 1$: $J_2(2) \oplus J_1(-1)^{\oplus 3}$. $\operatorname{tr} = 4$: $J_2(2) \oplus J_1(2) \oplus J_1(-1)^{\oplus 2}$. $\operatorname{tr} = 7$: $J_2(2) \oplus J_2(2) \oplus J_1(-1)$ (or $J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(-1)$). Each has largest block 2 for the eigenvalue 2 and only $1 \times 1$ blocks for $-1$, so $m_T = (x - 2)^2(x + 1)$.

**HW9 P3.**

(a) **Idea.** *Trace is the sum of the eigenvalues, and a nilpotent operator has only the eigenvalue 0.*
$N$ is nilpotent, so its only eigenvalue is 0 ($Nv = \lambda v$, $v \neq 0$ gives $0 = N^k v = \lambda^k v$). The trace is the sum of the eigenvalues with algebraic multiplicity, so $\operatorname{tr} N = 0$. (Or: in a Jordan basis $\mathcal{M}(N)$ has zero diagonal.)

(b) $\operatorname{diag}(1, -1)$: trace 0, and its square is $I$, so no power is 0.

(c) $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$: $AB = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right) \neq 0$, with trace $1 - 1 = 0$.

**Tr1.**

**Idea.** *$A^2 = -I$ pins down $m_A$; for a non-scalar $2 \times 2$ matrix, $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A$, so compare coefficients.*

$m_A \mid x^2 + 1$. $A = cI$ would need $c^2 = -1$ with $c$ real, impossible; so $\deg m_A = 2$ and $m_A = x^2 + 1$. For a non-scalar $2 \times 2$ matrix, $m_A = \chi_A = x^2 - (\operatorname{tr} A)x + \det A$. Comparing: $\operatorname{tr} A = 0$, $\det A = 1$.

**Tr2.**

**Idea.** *The exponents in $m_T$ are the largest blocks; fill the dimension 3.*

Algebraic multiplicities $a \geq 1$ (for 1) and $b \geq 2$ (for 2) with $a + b = 3$: $a = 1$, $b = 2$. So $T \sim J_1(1) \oplus J_2(2)$, $\chi_T = (x - 1)(x - 2)^2$, $\operatorname{tr} T = 1 + 2 + 2 = 5$.

## §2.4 Inner products (HW9)

**HW9 P4.**

**Idea.** *Linearity and symmetry are usually immediate; test positivity by looking for a nonzero $v$ with $\langle v, v\rangle \le 0$.*

$\langle x, y\rangle_1$: yes. It is linear in $x$, symmetric, and $\langle x, x\rangle_1 = x_1^2 + 2x_2^2 > 0$ for $x \neq 0$. $\langle x, y\rangle_2$: no. $\langle e_2, e_2\rangle_2 = -1 < 0$. $\langle x, y\rangle_3$: no. $\langle (1,-1),(1,-1)\rangle_3 = 0$ but $(1,-1) \neq 0$. (It is bilinear and symmetric; only positivity fails.)

**HW9 P5.**

(a) **Idea.** *Linearity and symmetry come from the integral; the content is positivity: a continuous function $\ge 0$ with integral $0$ is $0$.*
Linearity in $p$ comes from linearity of the integral; $\langle p, q\rangle = \langle q, p\rangle$; $\langle p, p\rangle = \int_{-1}^{1} p^2 \ge 0$. If $\int_{-1}^{1} p^2 = 0$, the continuous function $p^2 \ge 0$ is 0 on $[-1,1]$, so $p$ has infinitely many roots and $p = 0$.

(b) $\|1\|^2 = 2$, $\|x\|^2 = \int x^2 = \frac{2}{3}$, $\|x^2\|^2 = \int x^4 = \frac{2}{5}$: $\|1\| = \sqrt{2}$, $\|x\| = \sqrt{2/3}$, $\|x^2\| = \sqrt{2/5}$.

(c) $\langle 1, x\rangle = \int x = 0$ and $\langle x, x^2\rangle = \int x^3 = 0$ (odd integrands), but $\langle 1, x^2\rangle = \frac{2}{3} \neq 0$. So $1 \perp x$ and $x \perp x^2$; 1 and $x^2$ are not orthogonal.

**HW9 P6.**

**Idea.** *Write the left side as an inner product with the all-ones vector, then apply Cauchy–Schwarz.*

In $\mathbb{R}^3$ with the dot product, $a + b + c = \langle (a,b,c),(1,1,1)\rangle$, so $(a+b+c)^2 \le \|(a,b,c)\|^2 \|(1,1,1)\|^2 = 3(a^2+b^2+c^2)$. Equality holds iff $(a,b,c)$ and $(1,1,1)$ are dependent, i.e. iff $a = b = c$.

**HW9 P7.**

**Idea.** *Expand both squared norms; the cross terms appear with opposite signs.*

$\|u+v\|^2 = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle$ and $\|u-v\|^2 = \langle u,u\rangle - \langle u,v\rangle - \langle v,u\rangle + \langle v,v\rangle$. Adding, the cross terms cancel: $2\|u\|^2 + 2\|v\|^2$. This uses only (conjugate) bilinearity, so it holds over $\mathbb{R}$ and $\mathbb{C}$.

**I1.**

**Idea.** *Find two vectors whose inner product is the constant $n$ and whose squared norms are the two factors: $u = (\sqrt{a_i})$ and $v = (1/\sqrt{a_i})$.*

In $\mathbb{R}^n$, $\langle u, v\rangle = \sum \sqrt{a_i} \cdot \frac{1}{\sqrt{a_i}} = n$, $\|u\|^2 = \sum a_i$, $\|v\|^2 = \sum \frac{1}{a_i}$. Cauchy–Schwarz: $n^2 \le \left(\sum a_i\right)\left(\sum \frac{1}{a_i}\right)$. Equality iff $u = cv$, i.e. $\sqrt{a_i} = c/\sqrt{a_i}$, i.e. all $a_i$ equal $c$.

## §2.5 Orthogonality (HW10)

**HW10 P1.**

**Idea.** *Subtract shadows, then normalize. On $[-1,1]$ odd integrands vanish, which kills many of the inner products.*

$\|1\|^2 = 2$, so $e_1 = \frac{1}{\sqrt{2}}$. $x - \langle x, e_1\rangle e_1 = x - \frac{1}{2}\int_{-1}^{1} x\,dx = x$, and $\|x\| = \sqrt{2/3}$, so $e_2 = \sqrt{\frac{3}{2}}\,x$. $x^2 - \langle x^2, e_1\rangle e_1 - \langle x^2, e_2\rangle e_2 = x^2 - \frac{1}{2}\cdot\frac{2}{3} - 0 = x^2 - \frac{1}{3}$, with $\|x^2 - \frac{1}{3}\|^2 = \int_{-1}^{1}\left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right)dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$. So $e_3 = \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1)$.

Orthonormal basis: $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\frac{\sqrt{10}}{4}(3x^2 - 1)$.

**HW10 P2.**

**Idea.** *Perpendicular to a span means perpendicular to each spanning vector: a homogeneous linear system.*

$x \in W^\perp \iff x_1 - x_2 + x_3 = 0$ and $2x_2 + x_3 = 0$. Then $x_3 = -2x_2$, $x_1 = x_2 - x_3 = 3x_2$, and $x_4$ is free. Basis: $(3,1,-2,0)$, $(0,0,0,1)$. ($\dim W^\perp = 4 - 2$.)

**HW10 P3.**

**Idea.** *Same as Problem 2: impose $\langle p, 1+x\rangle = 0$ on a general $p$.*

For $p = a + bx + cx^2$: $\langle p, 1+x\rangle = \int_{-1}^{1}\left(a + (a+b)x + (b+c)x^2 + cx^3\right)dx = 2a + \frac{2}{3}(b+c)$. So $p \in W^\perp \iff 3a + b + c = 0$. Basis: $1 - 3x$ ($a = 1, c = 0$) and $x^2 - x$ ($a = 0, c = 1$). Check: $(1-3x)(1+x) = 1 - 2x - 3x^2$ integrates to $2 - 2 = 0$; $(x^2 - x)(1+x) = x^3 - x$ integrates to 0. ($\dim W^\perp = 3 - 1 = 2$.)

**HW10 P4.**

**Idea.** *Prove the easier identity for sums, then take complements of both sides; $(Z^\perp)^\perp = Z$ undoes each $\perp$.*

First, $(X+Y)^\perp = X^\perp \cap Y^\perp$ for any subspaces: $v \perp X + Y$ iff $v \perp X$ and $v \perp Y$ (every element of $X+Y$ is $x+y$, and $X, Y \subseteq X+Y$). Apply this to $X = U^\perp$, $Y = W^\perp$ and use $(Z^\perp)^\perp = Z$ (Prop. 90): $(U^\perp + W^\perp)^\perp = U \cap W$. Taking $\perp$ of both sides and using Prop. 90 again: $U^\perp + W^\perp = (U \cap W)^\perp$.

**HW10 P5.**

(a) **Idea.** *The $U^\perp$ part of $v$ is invisible to inner products with the $e_j$.*
Write $v = u + w$ with $u \in U$, $w \in U^\perp$ (Prop. 89); $P_U v = u$. Since $w \perp e_j$, $\langle v, e_j\rangle = \langle u, e_j\rangle$. As $e_1, \dots, e_m$ is an orthonormal basis of $U$, $u = \sum_j \langle u, e_j\rangle e_j = \sum_j \langle v, e_j\rangle e_j$.

(b) **Idea.** *Split the error into a part in $U^\perp$ that every $u$ must keep and a part in $U$ you can remove; Pythagoras adds them in squares.*
For $u \in U$, $v - u = (v - P_U v) + (P_U v - u)$ with $v - P_U v \in U^\perp$ and $P_U v - u \in U$. By Pythagoras $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \geq \|v - P_U v\|^2$. Equality iff $\|P_U v - u\| = 0$, i.e. iff $u = P_U v$.

**HW10 P6.**

**Idea.** *Riesz: find the representing vector by testing on an orthonormal basis; its $i$-th coordinate is $\psi(e_i)$ (no conjugates over $\mathbb{R}$).*

$\psi(f) = \int_0^2 f$ is a linear functional, so Riesz gives a unique $p$. Orthonormal basis of $\mathcal{P}_1(\mathbb{R})$: $e_1 = \frac{1}{\sqrt{2}}$, $e_2 = \sqrt{\frac{3}{2}}x$ (HW10 P1). $p = \psi(e_1)e_1 + \psi(e_2)e_2$ (real case, no conjugates). $\psi(e_1) = \frac{2}{\sqrt{2}} = \sqrt{2}$ and $\psi(e_2) = \sqrt{\frac{3}{2}} \cdot 2$. So $p = \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 2 \cdot \frac{3}{2}x = 1 + 3x$. Check: $\langle 1, 1 + 3x\rangle = 2 = \int_0^2 1$ and $\langle x, 1 + 3x\rangle = \int_{-1}^1 3x^2 = 2 = \int_0^2 x$.

**HW10 P7.**

**Idea.** *($\Leftarrow$) If some $u \in \operatorname{range} P$ and $w \in \operatorname{null} P$ were not perpendicular, moving $u$ slightly along $w$ would shorten it while $P$ still sends it to $u$, so $P$ would increase a length. To make this a proof, use $\|u + tw\|^2 \geq \|u\|^2$ for all $t$ and choose $t$ pointing against $\langle u, w\rangle$.*

($\Rightarrow$) If $P = P_U$, then $v = P_U v + (v - P_U v)$ with orthogonal pieces, so $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2 \geq \|P_U v\|^2$.
($\Leftarrow$) $V = \operatorname{null} P \oplus \operatorname{range} P$ (Prop. 91). Let $u \in \operatorname{range} P$, $w \in \operatorname{null} P$, and $t \in \mathbb{F}$. Then $P(u + tw) = u$ (as $Pu = u$, $Pw = 0$), so $\|u\|^2 \leq \|u + tw\|^2 = \|u\|^2 + 2\operatorname{Re}\left(\bar{t}\langle u, w\rangle\right) + |t|^2\|w\|^2$. Take $t = -s\langle u, w\rangle$ with $s > 0$ real: $0 \leq -2s|\langle u, w\rangle|^2 + s^2|\langle u, w\rangle|^2\|w\|^2$. Divide by $s$ and let $s \to 0^+$: $|\langle u, w\rangle|^2 \leq 0$, so $\langle u, w\rangle = 0$. Hence $\operatorname{null} P \subseteq (\operatorname{range} P)^\perp$; both have dimension $\dim V - \dim \operatorname{range} P$, so $\operatorname{null} P = (\operatorname{range} P)^\perp$. So for $v = u + w$ with $u \in \operatorname{range} P$, $w \in \operatorname{null} P = (\operatorname{range} P)^\perp$: $Pv = Pu + Pw = u = P_{\operatorname{range} P} v$; hence $P = P_{\operatorname{range} P}$.

**O1.**

**Idea.** *The integral is $\|x^2 - p\|^2$ for the $L^2$ inner product on $[0, 1]$, so the minimizer is the orthogonal projection of $x^2$ onto $\mathcal{P}_1(\mathbb{R})$ (best approximation).*

Orthonormal basis of $\mathcal{P}_1$ (Ex. 70): $e_1 = 1$, $e_2 = \sqrt{3}(2x - 1)$. $\langle x^2, e_1\rangle = \frac{1}{3}$, $\langle x^2, e_2\rangle = \sqrt{3}\int_0^1 x^2(2x - 1)\,dx = \frac{\sqrt{3}}{6}$. So $p = \frac{1}{3} + \frac{\sqrt{3}}{6} \cdot \sqrt{3}(2x - 1) = x - \frac{1}{6}$. Minimum: $\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx = \frac{1}{180}$ (this is $\|w_3\|^2$ in Ex. 70).

**O2.**

**Idea.** *Projection onto a line: $\frac{\langle v, u\rangle}{\|u\|^2}u$. $U^\perp$ is one linear equation.*

$P_U v = \frac{10}{4}(1, 1, 1, 1) = \left(\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, \frac{5}{2}\right)$. $U^\perp = \{x_1 + x_2 + x_3 + x_4 = 0\}$, basis $(1, -1, 0, 0), (0, 1, -1, 0), (0, 0, 1, -1)$. Distance $\|v - P_U v\| = \left\|\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\right)\right\| = \sqrt{5}$.

**O3.**

**Idea.** *Split $v = u + w$ along $U \oplus U^\perp$: $R$ keeps $w$ and flips $u$, a reflection.*

$Rv = v - 2P_U v = (u + w) - 2u = -u + w$, so $\|Rv\|^2 = \|u\|^2 + \|w\|^2 = \|v\|^2$ (Pythagoras). $R^2 = I - 4P_U + 4P_U^2 = I$ since $P_U^2 = P_U$.